Indian Institute of Engineering Science and Technology, Shibpur
B. Tech 4th Semester End-Semester Examination, May 2026

Computer Architecture and Organization-I (CS2202)

Full Marks: 50

Time: 3 Hours

Answer any five questions

1. (a) Consider the following reservation table of a 4-stage nonlinear pipeline.

	CC1	CC2	CC3	CC4	CC5	CC6
S1	X					X
S2		X		X		
S3			X			
S4				X	X	

(i) What are the forbidden latencies?
(ii) What is the initial collision vector?
(iii) Draw the pipeline.

(b) A pipeline has 4 segments with time delays 60 ns, 50 ns, 90 ns, 80 ns. Interface latch has delay 10 ns. Find the clock frequency of the pipeline.

(c) A non-pipeline system takes 50 ns to process a task. The same task can be processed in a 6 segment pipeline with a clock cycle time 10 ns. Find the speedup ratio of the pipeline for 100 tasks.

[(1.5+1.5+3)+2+2]

2. (a) Compare the relative merit of three cache memory organization (i) Direct-mapped (ii) Fully associative (iii) Set-associative

(b) Consider a direct mapped cache memory of size 32-kB with block size 32-byte. CPU generates 32-bit address. Find the number of bits needed for tag and cache indexing.

(c) The average access time is 140 ns without level and 30 ns with level of a two level memory. If the fastest memory access time is 20 ns, what is the hit ratio? [6+2+2]

3. (a) Define memory hierarchy.

(b) Define the following terms in the context of memory device characteristics. (i) Access time (ii) Access rate (iii) Access modes (iv) Cycle time. [2+(4x2)]

4. (a) What are the functions of IO interface circuit?

(b) What do you mean by bus control? In this context, discuss about bus master and bus slave.

(c) Compare 1-way and 2-way asynchronous data transfer. [2+(2+2+2)+2]

5. (a) What do you mean by bus arbitration?

(b) Discuss about the advantages and disadvantages of daisy chaining and polling bus arbitration. [2+4+4]

6. (a) Define addressing mode.

(b) Compare immediate addressing, direct addressing, indirect addressing in terms of the number of memory access for getting the operand.

(c) Compare the performance of Booth's multiplication algorithm and Bit-pair multiplication algorithm with example. [2+3+5]

7. Discuss about the following terms in the context of the control unit of the processor. (i) Microinstruction cycle (ii) Monophase mode of control (iii) Polyphase mode of control (iv) Control memory optimization (v) Microcode compaction [2x5]